

12.1 Kinematics of Uniform Circular Motion

Question Paper

Course	CIEA Level Physics
Section	12. Motion in a Circle
Topic	12.1 Kinematics of Uniform Circular Motion
Difficulty	Medium

Time allowed: 40

Score: /26

Percentage: /100

Question 1a

Explain why one complete revolution is equivalent to an angular displacement of 2π rad.

[1 mark]

Question 1b

A particle is following a circular path is observed to have an angular displacement of 0.22 rad.

Express this angle in degrees ($^{\circ}$). Show your working and give your answer to an appropriate number of significant figures.

[1 mark]

Question 1c

Calculate the percentage error that is made when the angle 0.22 rad, as measured in degrees (θ°), is assumed to be equal to $\tan(\theta^{\circ})$ to four significant figures. Give your final answer to two significant figures.

[2 marks]

Question 1d

The particle travels the angle in part (b) in 60 ms.

Calculate the frequency of the particle.

[3 marks]

Question 2a

State the difference between *angular velocity* and *linear velocity*.

[2 marks]

Question 2b

A moon is in a circular orbit r about a planet. The angular speed of the moon in its orbit is ω . The planet and its moon may be considered to be point masses that are isolated in space.

r and ω are related by the expression

$$r^3\omega^2 = \text{constant}$$

Io and Europa are moons that are in circular orbits around the planet Jupiter.

Data for Io and Europa are shown in Fig 1.1.

moon	radius of orbit / km	period of rotation around Jupiter / hours
Io	422 000	42.5
Europa	670 900	

Fig 1.1

Use the data from Fig 1.1. to determine the angular speed of Europa in its orbit around Jupiter.

[3 marks]

Question 2c

Calculate the time period of Europa.

 $T = \dots\dots\dots$ years

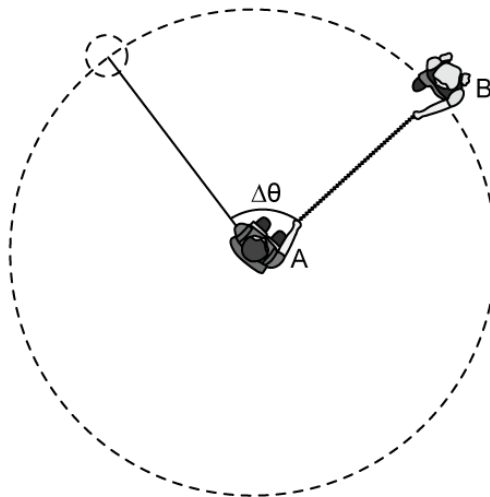
[2 marks]

Question 2d

Determine the radius of the orbit of a moon that had double the linear speed of Io but a third of its time period.

[2 marks]**Question 3a**

Two children A and B hold both sides of a rope 70 cm long. Child A swings around child B in a circular path, as shown Fig 1.1.

**Fig 1.1**

Child B travels a full rotation in 1.5 s.

Calculate the time taken for child B to travel a distance of 0.45 m.

[3 marks]

Question 3b

Show that the angular displacement, $\Delta\theta$ in this time is 37° .

[3 marks]**Question 3c**

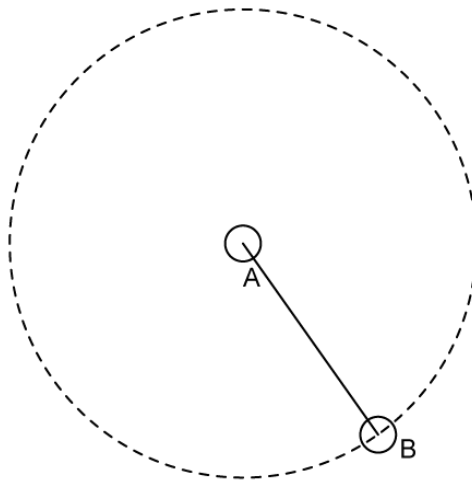
Child A now swings child B round till they are making 5 revolutions every 2 seconds.

Calculate the linear speed of child B.

[3 marks]

Question 3d

As the children are spinning anti-clockwise, the rope suddenly snaps at the position of child A and B shown in Fig 1.2.

**Fig 1.2**

Draw the direction child B will continue to travel in the instance the rope is snapped.

[1 mark]

Model Answers: Medium

1a

(a) One complete revolution is equivalent to an angular displacement of 2π rad because:

- Arc length is $r\theta$ so for one complete revolution arc = $2\pi r$ **AND** so $\theta = \frac{2\pi r}{r} = 2\pi$ [1 mark]

[Total: 1 mark]

1b

(b) Express this angle in degrees ($^{\circ}$)

List the known quantities:

- Angular displacement = 0.22 rad

State the equation converting from radians into degrees:

- $\theta(\text{rad}) = \theta(\text{degrees}) \times \left(\frac{\pi}{180}\right) \Rightarrow \theta(\text{degrees}) = \theta(\text{rad}) \times \left(\frac{180}{\pi}\right)$

Substitute in the values:

- $\theta^{\circ} = 0.22 \times \left(\frac{180}{\pi}\right) = 12.605^{\circ}$ [1 mark]

State your final answer to the appropriate number of significant figures:

- $\theta = 13^{\circ}$ (2 s.f) [1 mark]

[Total: 2 marks]

The appropriate number of significant figures is always the lowest number of the

numbers used in the calculation. In this case, 0.22 rad is used, which is to 2 s.f, so the final answer almost must be to 2 s.f.

1c

(c) Calculate the percentage error that is made when the angle 0.22 rad

List the known quantities:

- Angle 0.22 rads in degrees, $\theta^\circ = 12.605^\circ$ (from part b)

Calculate the value of $\tan(\theta^\circ)$:

- $\tan(12.605) = 0.2236$ (4 s.f.)

Calculate the percentage error:

- $\% \text{ error} = \frac{\text{difference in values}}{\text{original value}} \times 100\%$
- $\% \text{ error} = \frac{0.2236 - 0.22}{0.22} \times 100\%$ [1 mark]
- $\% \text{ error} = 1.636 = 1.6\%$ (2 s.f.) [1 mark]

[Total: 2 marks]

When calculating % differences, it is better to use the **unrounded** values from previous subparts, because the differences are often very small. The number of significant figures for the final answer asked for in the question is vital for the final mark.

Since 12.605 is the angle in **degrees** ensure your calculator is in **degrees** mode when using any trigonometric functions.

1d

(d) Calculate the angular velocity of the particle

List the known quantities:

- Time, $t = 60 \text{ ms} = 60 \times 10^{-3} \text{ s}$

Calculate the angular velocity of the particle:

- $\omega = \frac{\Delta\theta}{\Delta t} = \frac{0.22}{60 \times 10^{-3}} \text{ [1 mark]}$

- $\omega = 3.667 \text{ rad s}^{-1} \text{ [1 mark]}$

Calculate the frequency:

- $\omega = 2\pi f \Rightarrow f = \frac{\omega}{2\pi}$

- $f = \frac{3.667}{2\pi} = 0.58 \text{ Hz [1 mark]}$

[Total: 3 marks]

The angle θ in all equations in the circular motion topic **must** be in radians. Make sure not to incorrectly use your angle in degrees found in part (b) for this question.

2a

(a) The difference between angular velocity and linear velocity is:

- Angular velocity is the rate of change of angular displacement swept out by a radius; [1 mark]
- Whilst linear velocity is the rate of change (linear) displacement with respect to time when the object moves / for motion in a straight path; [1 mark]

[Total: 2 marks]

2b

(b) Use the data from Fig 1.1. the angular velocity of Europa in its orbit around Jupiter

List the known quantities:

- Radius of Io's orbit, $r_i = 422\,000 \text{ km} = 422\,000 \times 10^3 \text{ m}$
- Radius of Europe's orbit, $r_e = 670\,990 \text{ km} = 670\,990 \times 10^3 \text{ m}$
- Time period of Io's orbit, $T_i = 42.5 \text{ hours}$

Calculate the angular speed of Io, ω_i :

$$\bullet \omega = \frac{2\pi}{T} = \frac{2\pi}{42.5 \times 3600} = 4.1 \times 10^{-5} \text{ rad s}^{-1} \text{ [1 mark]}$$

Determine the equation for the angular speed of Europa, ω_e :

$$\bullet \text{ If } r^3 \omega^2 = \text{constant then } r_i^3 \omega_i^2 = r_e^3 \omega_e^2$$

$$\bullet \omega_e = \sqrt{\frac{r_i^3 \omega_i^2}{r_e^3}}$$

Substitute the values and calculate the angular speed, ω_e :

$$\bullet \omega_e = \sqrt{\frac{(420\,000 \times 10^3)^3 \times (4.1 \times 10^{-5})^2}{(670\,990 \times 10^3)^3}} \text{ [1 mark]}$$

$$\bullet \omega_e = 2.0 \times 10^{-5} \text{ rad s}^{-1} \text{ [1 mark]}$$

[Total: 3 marks]

Unless otherwise specified in the question, the angular speed is measured in rad s^{-1} .
This means the time period T must be in seconds (not hours, minutes or years).

2c

(c) Calculate the time period of Europa

List the known quantities:

$$\bullet \text{ The angular speed of Europa, } \omega = 2.0 \times 10^{-5} \text{ rad s}^{-1}$$

Calculate the time period in seconds:

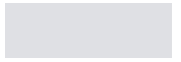
$$\bullet \omega = \frac{2\pi}{T} \Rightarrow T = \frac{2\pi}{\omega}$$

$$\bullet T = \frac{2\pi}{2.0 \times 10^{-5}} = 314.159 \times 10^3 \text{ s [1 mark]}$$

Convert the time period into years:

- $T = \frac{314.159 \times 10^3}{365 \times 24 \times 60 \times 60} = 9.96 \times 10^{-3} \text{ years}$ [1 mark]

[Total: 2 marks]



- $\omega_e = 2.0 \times 10^{-5} \text{ rad s}^{-1}$ [1 mark]

[Total: 3 marks]

Unless otherwise specified in the question, the angular speed is measured in rad s^{-1} . This means the time period T must be in seconds (not hours, minutes or years).

2c

(c) Calculate the time period of Europa

List the known quantities:

- The angular speed of Europa, $\omega = 2.0 \times 10^{-5} \text{ rad s}^{-1}$

Calculate the time period in seconds:

- $\omega = \frac{2\pi}{T} \Rightarrow T = \frac{2\pi}{\omega}$

- $T = \frac{2\pi}{2.0 \times 10^{-5}} = 314.159 \times 10^3 \text{ s}$ [1 mark]

Convert the time period into years:

- $T = \frac{314.159 \times 10^3}{365 \times 24 \times 60 \times 60} = 9.96 \times 10^{-3} \text{ years}$ [1 mark]

[Total: 2 marks]

2d

(d) The radius of orbit of a moon that had double the linear speed of Io but a third of the time period is:

List the known quantities:

- Radius of Io's orbit, $r_i = 422\,000 \text{ km}$

Determine the change in radius:

- $v = r\omega = \frac{2\pi r}{T}$
- $r = \frac{vT}{2\pi}$
- $r_{\text{moon}} = \frac{2v \times \frac{1}{3}T}{2\pi} = \frac{2}{3} \times \frac{vT}{2\pi} = \frac{2}{3}r_i$ [1 mark]

Calculate the radius of the moon, r_{moon} :

- $r_{\text{moon}} = \frac{2}{3} \times (422\,000) = 281\,333 = 280\,000 \text{ km}$ [1 mark]

[Total: 2 marks]

The final answer in metres will also be accepted.

Questions like this are also very common in multiple-choice questions. You must first find the equation that links all the variables you are looking for (in this case, orbital radius, linear speed and time period) and compare them from there.

If the linear speed was doubled and the time period was **halved** instead, you can see that $2 \times \frac{1}{2}$ will cancel out and actually the orbital radius would be the same. Watch out for this as well.

3a

(a) Calculate the time taken for child B to travel a distance of 0.45 m

List the known quantities:

- Distance, $\Delta s = 0.45 \text{ m}$
- Time period, $T = 1.5 \text{ s}$
- Radius, $r = 70 \text{ cm} = 0.70 \text{ m}$

Calculate the angular velocity, ω :

- $\omega = \frac{2\pi}{T} = \frac{2\pi}{1.5} = 4.189 \text{ rad s}^{-1}$ [1 mark]

Calculate the time taken:

- $\Delta\theta = \omega\Delta t = \frac{\Delta s}{r}$
- $\Delta t = \frac{\Delta s}{r\omega} = \frac{0.45}{0.70 \times 4.189}$ [1 mark]
- $\Delta t = 0.1545 = 0.15 \text{ s}$ [1 mark]

[Total: 3 marks]

None of the circular motion equations will be given on your data sheet, so make sure you remember these, and be confident in rearranging them.

3b

(b) Show that the angular displacement, $\Delta\theta$ in this time is 37°

List the known quantities:

- Angular velocity, $\omega = 4.189 \text{ rad s}^{-1}$
- Time, $\Delta t = 0.1545 \text{ s}$

Calculate the angular displacement, $\Delta\theta$:

- $\omega = \frac{\Delta\theta}{\Delta t} \Rightarrow \Delta\theta = \omega\Delta t$
- $\Delta\theta = 4.189 \times 0.1545$ [1 mark]
- $\Delta\theta = 0.647 \text{ rad}$ [1 mark]

Convert $\Delta\theta$ from radians to degrees:

- $\theta (\text{degrees}) = \theta (\text{rad}) \times \left(\frac{180}{\pi}\right) = 0.647 \times \left(\frac{180}{\pi}\right)$
- $\Delta\theta = 37^\circ$ [1 mark]

[Total: 3 marks]

3c

(c) Calculate the linear speed of child B

List the known quantities:

- Radius, $r = 70 \text{ cm} = 0.70 \text{ m}$

Calculate the frequency, f :

- Frequency is the number of revolutions in 1 second
- 5 revolutions in 2 seconds so f is $\frac{5}{2} = 2.5$ revolutions per second (2.5 s^{-1}) [1 mark]

Calculate the linear speed, v :

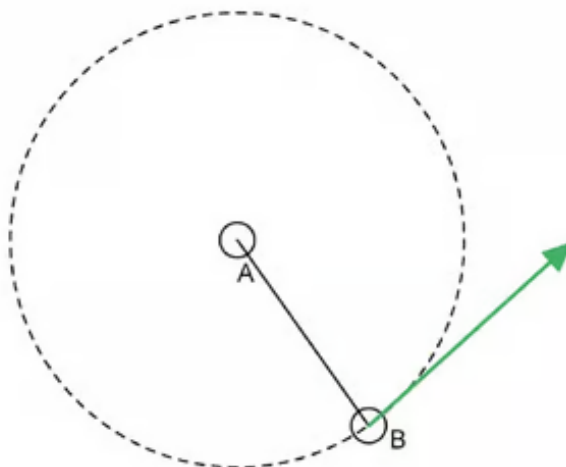
- $v = \omega r = (2\pi f)r$
- $v = 2\pi \times 2.5 \times 0.70$ [1 mark]
- $v = 10.996 = 11 \text{ m s}^{-1}$ [1 mark]

[Total: 3 marks]

3d

(d) Sketch on the diagram the direction child B will continue to travel in the instance the rope is snapped.

A sketch that would score 1 mark should look like:



- Arrow tangent to be pointed anti-clockwise; [1 mark]

[Total: 1 mark]

